TasteRank

Eigenvector Centrality on WSET Systematic Approach to Tasting Profiles

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Methodology and Technical Documentation

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Working Paper — Companion to Wine Grape Similarity Network

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1. Introduction and Motivation

This document presents TasteRank, a centrality measure for wine grape varieties derived entirely from empirical tasting characteristics as codified in the WSET Systematic Approach to Tasting (SAT). Where the companion Wine Grape Similarity Network relied on expert-curated similarity judgments, TasteRank constructs similarity computationally from a structured feature space, offering an independent and reproducible measure of varietal centrality.

The core question is: which grape varieties occupy the most structurally central positions in the tasting universe, when centrality is measured not by reputation, acreage, or critical opinion, but by proximity in a rigorously defined multidimensional flavor space?

The approach has three stages. **First**, each of 101 grape varieties is encoded as a 13-dimensional vector capturing appearance, nose, and palate characteristics. **Second**, pairwise cosine similarity between all vectors is computed, and a k-nearest-neighbor graph is constructed ($k = 5$). **Third**, eigenvector centrality is computed on the resulting weighted graph, yielding a single scalar TasteRank score per variety.

This methodology offers several advantages over the expert-curated approach: it is fully reproducible given the input vectors; similarity is continuous (weighted edges) rather than binary; and it is extensible to any variety for which a SAT profile can be specified.

2. Feature Space Construction

2.1 The WSET Systematic Approach to Tasting

The WSET SAT provides a standardized vocabulary and assessment framework for evaluating wine across three sensory domains: appearance, nose, and palate. We formalize this framework into a 13-dimensional feature vector. Each dimension is scaled on an ordinal scale, with most dimensions ranging from 1 (low/absent) to 5 (high/pronounced). For tannin, white varieties are assigned a value of 0 (structurally absent), while reds range from 1 to 5. For aromatic sub-dimensions (floral, herbal/earthy, spice/oak), the scale runs from 0 (absent) to 5 (defining characteristic).

2.2 Feature Dimensions

#	Dimension	Domain	Scale	Interpretation
1	Color Depth	Appearance	1–5	Pale → Deep; visual proxy for extraction, skin contact, tannin potential
2	Aromatic Intensity	Nose	1–5	Light → Pronounced; overall olfactory impact
3	Floral	Nose	0–5	Absent → Defining; rose, violet, blossom, linalool-driven aromas
4	Fruit Ripeness	Nose/Palate	1–5	Green/tart → Dried/compote; position on the ripeness spectrum
5	Herbal/Earthy	Nose/Palate	0–5	Absent → Defining; pyrazines, garrigue, sous-bois, mineral
6	Spice/Oak	Nose/Palate	0–5	Absent → Defining; pepper, clove, cedar, vanilla, toast
7	Acidity	Palate	1–5	Low → High; structural backbone, freshness
8	Tannin	Palate	0–5	None (white) → High; phenolic grip and astringency
9	Body	Palate	1–5	Light → Full; perceived weight on palate
10	Alcohol	Palate	1–5	Low → High; perceived warmth, viscosity contribution
11	Flavor Intensity	Palate	1–5	Light → Pronounced; overall palate flavor concentration
12	Finish Length	Palate	1–5	Short → Long; persistence of flavor after swallowing
13	Complexity	Palate	1–5	Simple → Complex; layered flavors, evolution in glass

2.3 Encoding Principles

Each grape variety is encoded at its *typical* expression: the central tendency of wines made from that variety under standard commercial vinification, at moderate quality levels. Extreme expressions (e.g., late-harvest Riesling, carbonic maceration Gamay, barrique-aged Vermentino) are not encoded because they represent winemaking choices rather than varietal character. Where regional variation is significant (e.g., cool-climate vs. warm-climate Syrah), the encoding reflects the global average weighted toward the variety's principal growing regions.

The encoding draws on the same three reference sources as the companion study: the WSET Level 3 and Diploma study materials (for SAT calibration), Wine Grapes (Robinson, Harding, Vouillamoz, 2012) for varietal typicity, and Wine Folly's tasting profiles for accessible cross-variety comparisons.

2.4 Dataset Scope

The dataset comprises 101 grape varieties (53 red, 48 white), identical in scope to the companion similarity network study. Each variety is represented by a single 13-dimensional vector. The complete dataset constitutes a 101 × 13 feature matrix X.

3. Analytical Framework

3.1 Cosine Similarity

Pairwise similarity between grape varieties is computed using cosine similarity, which measures the angular proximity of two vectors in the 13-dimensional feature space:

$$cos(\theta_{ij}) = (x_i \cdot x_j) / (||x_i|| \cdot ||x_j||)$$

Cosine similarity is preferred over Euclidean distance for two reasons. First, it is scale-invariant: two varieties with proportionally similar profiles but different absolute magnitudes (e.g., a light variety scoring 1-2 across all dimensions vs. a full variety scoring 4-5) will have high cosine similarity, correctly capturing their structural parallel. Second, cosine similarity naturally handles the zero values in the tannin dimension for white varieties, whereas Euclidean distance would penalize red-white pairs excessively for this single structural difference.

The result is a 101 × 101 symmetric similarity matrix S where $S_{ij} = \cos(\theta_{ij})$. Diagonal entries are set to zero (a variety is not its own neighbor). Empirically, the similarity values are very high: the mean pairwise cosine similarity among the k-nearest neighbors is 0.983, the median is 0.986, and the minimum edge weight in the graph is 0.909.

3.2 k-Nearest-Neighbor Graph Construction

From the similarity matrix, we construct a k-nearest-neighbor (kNN) graph with k = 5. For each variety i, we identify the 5 varieties with the highest cosine similarity to i and create weighted edges between them, with edge weight equal to the cosine similarity score.

The graph is symmetrized: if variety j is among the 5 nearest neighbors of i, or i is among the 5 nearest neighbors of j, the edge (i, j) exists. Where both directions produce an edge, the maximum of the two weights is retained. This symmetrization typically produces a graph with average degree slightly above k, since not all nearest-neighbor relationships are reciprocal.

Metric	Value
Nodes (varieties)	101
Edges (similarity links)	341
Average degree	6.75
Mean edge weight (cosine sim.)	0.9829
Median edge weight	0.9860
Min edge weight	0.9092
Max edge weight	1.0000
Communities detected	6

3.3 Eigenvector Centrality (TasteRank)

TasteRank is defined as the eigenvector centrality of each node in the weighted kNN graph. For a weighted adjacency matrix W (where W_{ij} = cosine similarity if an edge exists, 0 otherwise), the eigenvector centrality x satisfies:

$$W \cdot x = \lambda_1 \cdot x$$

where λ_1 is the largest eigenvalue of W and x_i is the TasteRank score of variety i. The use of weighted edges means that a variety connected to neighbors with high cosine similarity scores earns more centrality than one connected to more distant neighbors, even if their degree is identical.

The interpretation is direct: a grape variety has high TasteRank if its SAT profile is similar to many other varieties that themselves have similar profiles to many others. High TasteRank identifies the varieties that sit at the dense center of the tasting universe—the “generic” archetypes around which other varieties cluster.

3.4 PageRank as Robustness Check

As in the companion study, we compute PageRank ($\alpha = 0.85$) on the same weighted graph as a robustness check. The damping factor redistributes centrality more evenly, reducing the dominance of the densest subgraph. Agreement between TasteRank and PageRank validates that the centrality structure is robust to the choice of spectral measure.

3.5 Community Detection

Community structure is detected using the Clauset–Newman–Moore greedy modularity optimization algorithm, applied to the weighted graph. The algorithm identifies groups of varieties that are more densely connected to each other (in cosine similarity space) than to the rest of the network.

4. Results

4.1 TasteRank Rankings

The top 20 varieties by TasteRank are presented below, alongside their PageRank scores and node degree.

Rank	Variety	Type	TasteRk	PageRk	Degree	Avg wt
1	**Sagrantino**	Red	0.3020	0.0146	11	0.983
2	**Nero d'Avola**	Red	0.2849	0.0134	10	0.985
3	**Lagrein**	Red	0.2748	0.0148	11	0.982
4	**Negroamaro**	Red	0.2380	0.0136	10	0.981
5	**Plavac Mali**	Red	0.2251	0.0110	8	0.984
6	**Montepulciano**	Red	0.2108	0.0098	7	0.985
7	**Petit Verdot**	Red	0.2093	0.0125	9	0.982
8	**Monastrell**	Red	0.2060	0.0111	8	0.983
9	**Petite Sirah**	Red	0.1768	0.0086	6	0.986
10	**Pinotage**	Red	0.1710	0.0086	6	0.985
11	**Malbec**	Red	0.1697	0.0086	6	0.986
12	**Tannat**	Red	0.1676	0.0086	6	0.985
13	**Garnacha Tint.**	Red	0.1670	0.0073	5	0.987
14	**Dolcetto**	Red	0.1659	0.0086	6	0.981
15	**Mourvèdre**	Red	0.1565	0.0113	8	0.980
16	**Zinfandel**	Red	0.1498	0.0087	6	0.984
17	**Primitivo**	Red	0.1498	0.0087	6	0.984
18	**Merlot**	Red	0.1333	0.0087	6	0.980
19	**Carignan**	Red	0.1283	0.0100	7	0.978
20	**Limniona**	Red	0.1223	0.0140	10	0.975

4.2 Community Structure

The modularity optimization algorithm detects 6 communities, compared to 5 in the expert-curated network. The additional community arises from the finer resolution of cosine similarity in the white grape space.

ID	Character	Size	R / W	Hub Varieties
C0	Full-bodied Mediterranean/Italian Reds	31	31 / 0	Sagrantino, Nero d'Avola, Lagrein, Negroamaro

C1	Light/Aromatic Cross-boundary	21	4 / 17	Assyrtiko, Falanghina, Vidiano, Ribolla Gialla
C2	Mid-weight Structured Reds	18	18 / 0	St. Laurent, Zweigelt, Counoise, Blaufränkisch
C3	Rich Textural Whites	13	0 / 13	Marsanne, Fiano, Godello, Pinot Gris
C4	Mineral/Crisp Medium Whites	10	0 / 10	Greco, Friulano, Verdicchio, Verdejo
C5	Lean Neutral Whites	8	0 / 8	Mauzac, Cortese, Aligoté, Melon de Bourgogne

The most structurally significant finding is Community C1, which contains both red and white varieties. Its 4 red members—Schiava, Poulsard, Frappato, and Kadarka—are all very light-bodied reds with minimal tannin (scores of 1–2), positioning them closer in SAT space to aromatic, medium-bodied whites than to their nominal red peers. This algorithmically confirms what sommeliers observe experientially: these varieties inhabit a liminal zone where the red-white distinction becomes secondary to structural and aromatic character.

5. Comparative Analysis: TasteRank vs. Expert Similarity Network

5.1 Divergent Rankings

The two methodologies produce substantially different centrality rankings. The following table compares the top 10 from each approach.

Rank	Expert Network	Eigen.	Rank	TasteRank	TasteRk
1	**Monastrell**	0.360	1	**Sagrantino**	0.302
2	**Mourvèdre**	0.315	2	**Nero d'Avola**	0.285
3	**Syrah**	0.294	3	**Lagrein**	0.275
4	**Tannat**	0.268	4	**Negroamaro**	0.238
5	**Petite Sirah**	0.203	5	**Plavac Mali**	0.225
6	**Touriga Nacional**	0.203	6	**Montepulciano**	0.211
7	**Grenache**	0.200	7	**Petit Verdot**	0.209
8	**Malbec**	0.187	8	**Monastrell**	0.206
9	**Carignan**	0.187	9	**Petite Sirah**	0.177
10	**Nero d'Avola**	0.180	10	**Pinotage**	0.171

5.2 Interpretation of Divergences

Sagrantino rises from unranked (#20) to #1. In the expert network, Sagrantino is connected only to peers in the austere/tannic niche (Tannat, Aglianico, Petit Verdot, Mourvèdre). Its expert similarity set is narrow. But its SAT vector—deep color, high tannin, full body, high alcohol, moderate spice—places it at the geometric centroid of the largest red cluster. It is the most "average" among full-bodied reds, which is precisely what eigenvector centrality rewards in a cosine-similarity graph.

Touriga Nacional drops from #6 to ~#33. Touriga Nacional's SAT profile is distinctive: it scores 5 on aromatic intensity and floral (one of very few reds with pronounced floral character), plus 5 on complexity and finish. This distinctive signature makes it an outlier in cosine space—few varieties are similar to it. Expert judgment recognizes its quality connections to Cabernet Sauvignon and Mourvèdre, but the tasting profile sits in a sparsely populated region of the feature space.

Cabernet Sauvignon drops from #11 to #30. Its combination of pronounced herbal/earthy character (3), high spice/oak (4), and maximum tannin (5) with only moderate fruit ripeness (3) creates a distinctive vector. High distinctiveness → low cosine centrality. The same logic explains Nebbiolo's poor showing: its combination of high acidity (5), high tannin (5), medium body (3), and pronounced aromatics is nearly unique.

Syrah drops from #3 to #24. Syrah's profile is differentiated by its high spice/oak dimension (4) and pronounced complexity (5). While experts recognize Syrah's broad stylistic family (Mourvèdre, Monastrell, Malbec, Petite Sirah), the SAT encoding captures its distinctiveness more than its typicality. This reveals a fundamental tension between the two approaches: experts encode "what tastes like what" incorporating context, while cosine similarity encodes strict profile overlap.

5.3 The Centrality–Distinctiveness Tradeoff

The divergence between the two rankings can be formalized as a tradeoff between centrality and distinctiveness. Define the distinctiveness of variety i as:

$$D_i = 1 - (1/n) \cdot \Sigma_j \cos(\theta_{ij})$$

A variety with high distinctiveness has low average cosine similarity to the rest of the universe. In the expert network, distinctiveness does not penalize centrality because experts can assign cross-cluster bridges based on contextual knowledge (e.g., "Syrah resembles Mourvèdre in the Rhône blend context"). In the TasteRank framework, distinctiveness is mechanically penalized because distinctive profiles have fewer high-similarity neighbors.

This tradeoff explains why the expert network rewards bridge grapes (Monastrell, Tannat) that connect stylistically distant clusters, while TasteRank rewards centroid grapes (Sagrantino, Nero d'Avola, Lagrein) that sit in the densest region of the feature space. Both perspectives are valid: bridge centrality measures structural importance, while centroid centrality measures typicality.

6. The Red–White Boundary in Tasting Space

A major finding of the TasteRank analysis is the partial dissolution of the red-white boundary in the cosine similarity graph. While the expert-curated network produces perfect red-white separation across all communities, the SAT-based network identifies a mixed community (C1) containing four light reds and seventeen whites.

6.1 Cross-Color Similarity Pairs

The following pairs exhibit cosine similarity above 0.95, indicating near-identical SAT profiles despite different grape colors. These are ordered by similarity score.

Red Variety	White Variety	Cosine	Shared Profile
Frappato	Loureiro	0.981	Light body, floral, high acid, minimal tannin, short finish
Gamay	Ribolla Gialla	0.970	Light body, moderate aromatics, high acid, low tannin
Gamay	Fiano	0.969	Medium aromatics, moderate fruit, high acid, light-medium body
Gamay	Petit Manseng	0.968	High acid, aromatic, moderate body, notable complexity
Corvina	Vidiano	0.971	Medium everything: body, acid, aromatics, complexity
Schiava	Welschriesling	0.972	Pale, low tannin, moderate acid, light body, simple
Schiava	Albariño	0.971	Pale/light, high acid, low body, floral hints
Grenache	Marsanne	0.973	Low acid, full body, high alcohol, moderate intensity
Zweigelt	Pinot Gris	0.964	Medium across all dimensions, balanced, understated

6.2 Structural Interpretation

The cross-color pairs cluster into two types. The first is the light-red/aromatic-white convergence: Frappato, Gamay, Schiava, and Poulsard have SAT profiles dominated by high acidity, low tannin, and light body—the dimensions they share with structured whites like Albariño, Fiano, and Loureiro. The tannin dimension (the primary red-white separator) is effectively neutralized because these reds score only 1–2, while the other 12 dimensions drive similarity.

The second type is the full-bodied convergence: Grenache–Marsanne, Merlot–Chardonnay, and Primitivo–Marsanne pairs share low acidity, full body, high alcohol, and moderate complexity.

Here the convergence is driven by palate structure (body, alcohol, finish) overwhelming the tannin differential.

These findings suggest that the red-white divide, while structurally dominant in the full graph, is not absolute. It is strongest in the mid-range—where tannin levels of 3–4 create clear separation—and weakest at the extremes (very light reds, very full whites).

7. Visualization

The TasteRank network is visualized using the same Fruchterman–Reingold force-directed layout as the companion study, with several modifications to improve readability.

Weighted layout. Edge weights (cosine similarity scores) are passed to the spring layout algorithm, causing varieties with higher similarity to be attracted more strongly. This produces tighter clustering of genuinely similar varieties compared to the unweighted layout.

Visual encoding. Node size is linearly scaled by TasteRank (range 250–4,000 units). Node color reflects community membership with distinct hues per community (warm tones for red communities, cool/gold tones for white communities). Edge width and opacity are both scaled by cosine similarity weight, with minimum opacity of 0.15 and maximum of 0.60, making stronger connections visually prominent. The background uses a warm parchment tone (#F5F0EB) for improved legibility, and edge colors are rendered at higher saturation than in the companion study.

Label hierarchy. Font size is scaled by TasteRank (range 6.5–15.5pt), and the top 15 varieties are rendered in bold. The top 35 subgraph includes rank annotations.

8. Limitations and Extensions

8.1 Encoding Subjectivity

While the analytical pipeline is fully reproducible given the input vectors, the encoding of SAT profiles involves expert judgment. Two experienced tasters might disagree on whether Nebbiolo's aromatic intensity is 4 or 5, or whether Chenin Blanc's typical body is 3 or 2. Sensitivity analysis—perturbing each dimension by ±1 and recomputing TasteRank—would quantify the stability of the rankings under encoding uncertainty.

8.2 Ordinal vs. Ratio Scales

The SAT dimensions are ordinal (the difference between acidity 3 and 4 is not necessarily equal to the difference between 4 and 5), but cosine similarity treats them as ratio-scale variables. This is a standard approximation in sensory science but introduces measurement imprecision. An alternative would be to use ordinal distance measures (e.g., Kendall's tau or Spearman correlation) instead of cosine similarity.

8.3 Dimensionality and Weighting

All 13 dimensions receive equal weight in the cosine computation. In practice, tannin and acidity may be more discriminating than color depth or alcohol for varietal identification. A principal component analysis (PCA) or expert-weighted cosine metric could better reflect the relative importance of each dimension. Preliminary analysis suggests that the first two principal components explain approximately 55% of variance, with the first PC dominated by the body–tannin–alcohol axis and the second by the acidity–complexity–finish axis.

8.4 Potential Extensions

Hybrid TasteRank. A natural synthesis would combine the SAT-based cosine similarity (empirical, reproducible) with the expert-curated edges (contextual, bridge-aware) into a single weighted graph. Edge weights could be a convex combination: $w_{i\square} = \alpha \cdot \cos(\theta_{i\square}) + (1 - \alpha) \cdot \text{Expert}_{i\square}$, where α is a tuning parameter.

Dynamic profiling. SAT profiles could be encoded at multiple price/quality tiers (entry, mid, premium), recognizing that the same variety often expresses differently at different quality levels. This would transform each grape from a single point to a trajectory in the 13-dimensional space.

Integration with The Soul of Wine. The TasteRank framework could be extended to a region-level analysis by aggregating the SAT profiles of each region's principal varieties, weighted by planting density. This would produce a "RegionRank" measuring which wine regions occupy the most central positions in the global tasting landscape—a quantitative companion to the metaphorical framework of The Soul of Wine.

9. Implementation

The analysis is implemented in Python 3.x using NetworkX 3.x (graph construction, eigenvector centrality, PageRank, community detection), NumPy (vectorized cosine similarity computation, kNN selection), and Matplotlib 3.x (Fruchterman–Reingold layout, network visualization).

The Fruchterman–Reingold layout uses parameters k = 2.0 (optimal distance), 250 iterations, weight-aware spring forces, and a fixed random seed (42) for reproducibility. Eigenvector centrality is computed with tolerance 10^{-6} and maximum 2,000 iterations. The kNN graph uses k = 5 with maximum-weight symmetrization.

The complete source code, including the 101 × 13 SAT feature matrix, is available as a standalone Python script for reproduction and extension.

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